Utilizing NMR Methods to Determine the KETO-ENOL Equilibrium

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INTRODUCTION

Nuclear Magnetic Resonance (NMR) spectroscopy is an analytical technique that utilizes the magnetic properties of atomic nuclei to measure information about the molecular structure, chemical environment, and physical dynamics. This method of spectroscopy has become a standard across the physical sciences, from studying the composition of foods to drug development. The fundamental principle of NMR involves the absorption of radiofrequency radiation by the nuclei when placed in a strong magnetic field. The exact frequency depends on the nuclei's electronic environment. This sensitivity makes NMR valuable for the quantification and identification of different chemical species in a mixture.

One prominent application of NMR spectroscopy is pharmaceutical drug design and discovery. It has been hailed as the "Gold Standard Method" for characterizing molecular structures, providing a window into studying drug-target interactions (Emwas et al., 2020). NMR allows researchers to monitor chemical reactions, assess the purity of pharmaceutical compounds, and determine the structure of drug candidates. For this experiment, proton NMR (^1H NMR) was used for the investigation of keto-enol tautomerism in β-dicarbonyl compounds. By analyzing the intensities of the signals from both the keto and enol forms, the change in Gibbs free energy and equilibrium constant for the tautomerization process were determined.

THEORY

Nuclear magnetic resonance works by detecting how atomic nuclei respond to being placed in a strong magnetic field. At specific frequencies, these nuclei will absorb radiofrequency radiation depending on their chemical environment. This is expressed as chemical shift (δ), which is how far a signal appears from a reference standard. In this experiment, chemical shifts were calculated relative to tetramethylsilane (TMS) using equation 1.

Equation 1: Chemical Shift

$$\delta_i = \frac{(H_r - H_i)}{H_r} * 10^6$$

In this equation, δ_i is the chemical shift in parts per million (ppm), H_r is the applied resonant field for the reference (TMS), and H_i is the applied resonant field for the nucleus being measured. Depending on the chemical environment, protons will show up at different positions on the spectrum. The left side of the spectrum is often referred to as "downfield"; this is the direction that electron-withdrawing groups resonate at, as they pull electron density from nearby protons. Electron-donating groups appear on the right side of the spectrum, as they have the opposite effect.

Through integrating the NMR peak, the area is proportional to the number of protons that produce the signal. This integrated area is relative to the other values in the signal; if one signal has a value of 3.0 and another has a value of 1.0, this could represent 3 protons and 1 proton, or 6 and 2, or 9 and 3, aligning with a 3:1 ratio. The integrated area was used to determine how much of each tautomer was present in the solution. β-dicarbonyl compounds exist in two forms: a keto form that has a carbonyl group, and an enol form that has a double bond and a hydroxyl group. These structures interconvert through tautomerization, where a proton shifts from a carbon next to the carbonyl to the oxygen. The enol form is stabilized through hydrogen bonding between the hydroxyl hydrogen and the carbonyl oxygen, which is why both forms can exist at the same time.

The equilibrium constant (K_{eq}) can be calculated using equation 2.

Equation 2:

$$K_{eq} = \frac{[enol]}{[keto]}$$

In this equation, [enol] and [keto] are the molar concentrations of each tautomer. K_{eq} is the equilibrium constant. The ratio of the integrated peaks can be substituted for the concentration as the ratio is proportional. This is only the case due to the same number of protons being present; if the proton count were different, a correction factor would need to be applied.

Lastly, the change in Gibbs free energy ($\Delta G°$) can be calculated using equation 3.

Equation 3:

$$\Delta G° =- RTln(K_{eq})$$

In this equation, R is the gas constant (8.314 J $mol^{-1}K^{-1}$), T is the temperature in Kelvin, and $ln(K_{eq})$ is the natural log of the equilibrium constant calculated in equation 2. If $\Delta G°$ is negative, this indicates the enol is favored, with a positive indicating the keto form is more stable.

EXPERIMENTAL

Chemicals and Equipment

The three analyzed chemicals for this experiment were acetylacetone ($C_5H_8O_2$), ethyl acetate ($C_4H_8O_2$), and ethyl acetoacetate ($C_6H_{10}O_3$). They were dissolved in deuterated chloroform ($CDCL_3$). The deuterated chloroform served as the NMR solvent due to its lack of interfering proton signals; the deuterium nuclei do not appear on the ^{1}H NMR spectrum. The measurements were taken on a 400 MHz spectrometer (Bruker), in 5 mm NMR sample tubes. The NMR spectrometer was able to calculate the integration and measure the relative peaks.

Experimental Procedure

For each solution, a 5 mm NMR tube was filled approximately 5 cm high with deuterated chloroform. A pipette was used to add three drops of the compounds to the NMR tube. The solution was mixed by drawing the liquid into the pipette and dispensing it. The three solutions contained ethyl acetate, acetylacetone, and ethyl acetoacetate dissolved in $CDCl_3$. The NMR spectra were measured using the 400 MHz Bruker spectrometer. The spectra were provided through email, and the values for each peak were included.

DATA

The NMR spectra were acquired at approximately room temperature (298 Kelvin). The chemical shifts are reported in parts per million (ppm) relative to the residual $CDCl_3$ peak. Since ethyl acetate has no keto-enol equilibrium, no spectrum is reported.

Figure 1: NMR spectra of acetylacetone

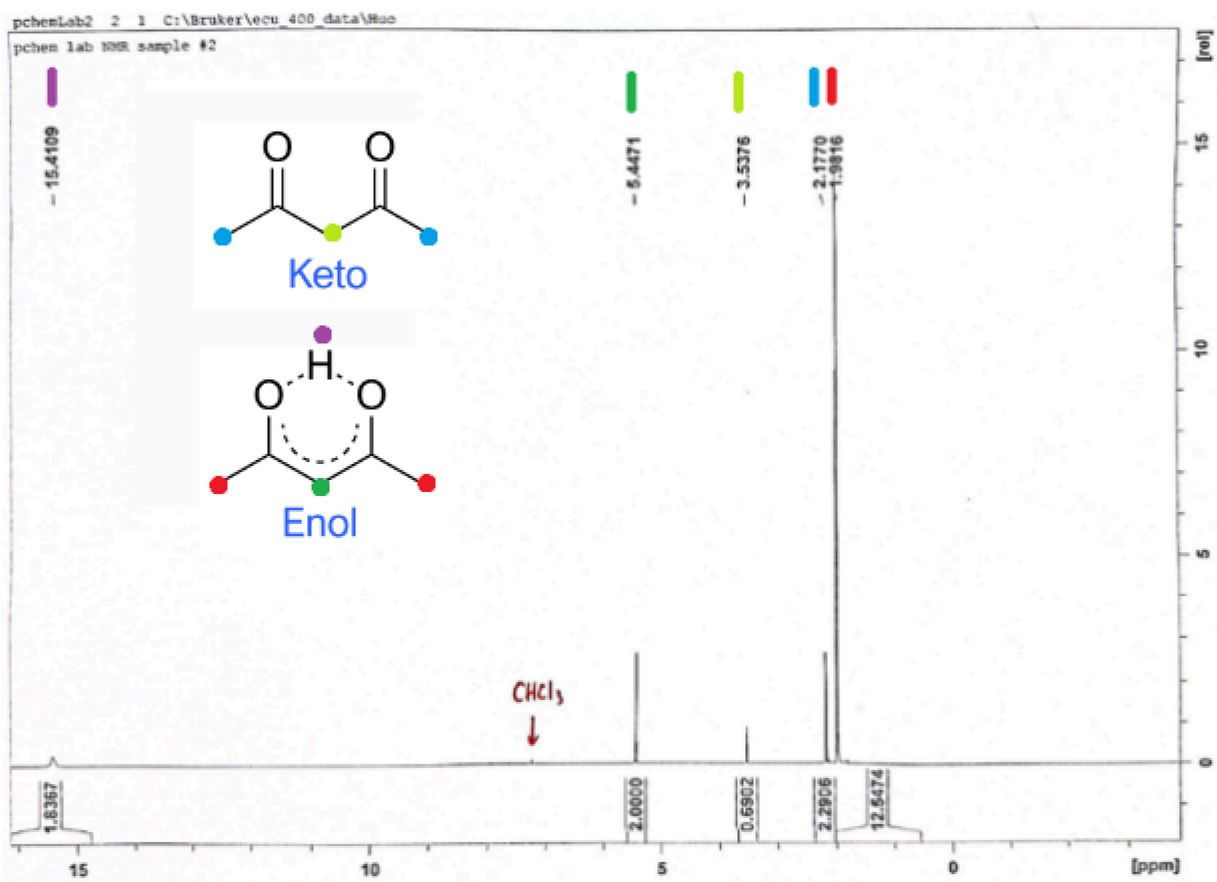

Table 1: NMR data for acetylacetone

Chemical Shift (ppm)	Integration Value	Splitting Pattern	Compound Form	Proton Assignment
1.98	12.55	Singlet	Enol	$-CH_3$
2.18	2.30	Singlet	Keto	$-CH_3$
3.54	0.69	Singlet	Keto	$=CH-$
5.45	2.00	Singlet	Enol	$-CH_2-$
15.41	1.84	Singlet	Enol	$-OH$

Figure 2: NMR spectra for ethyl acetoacetate

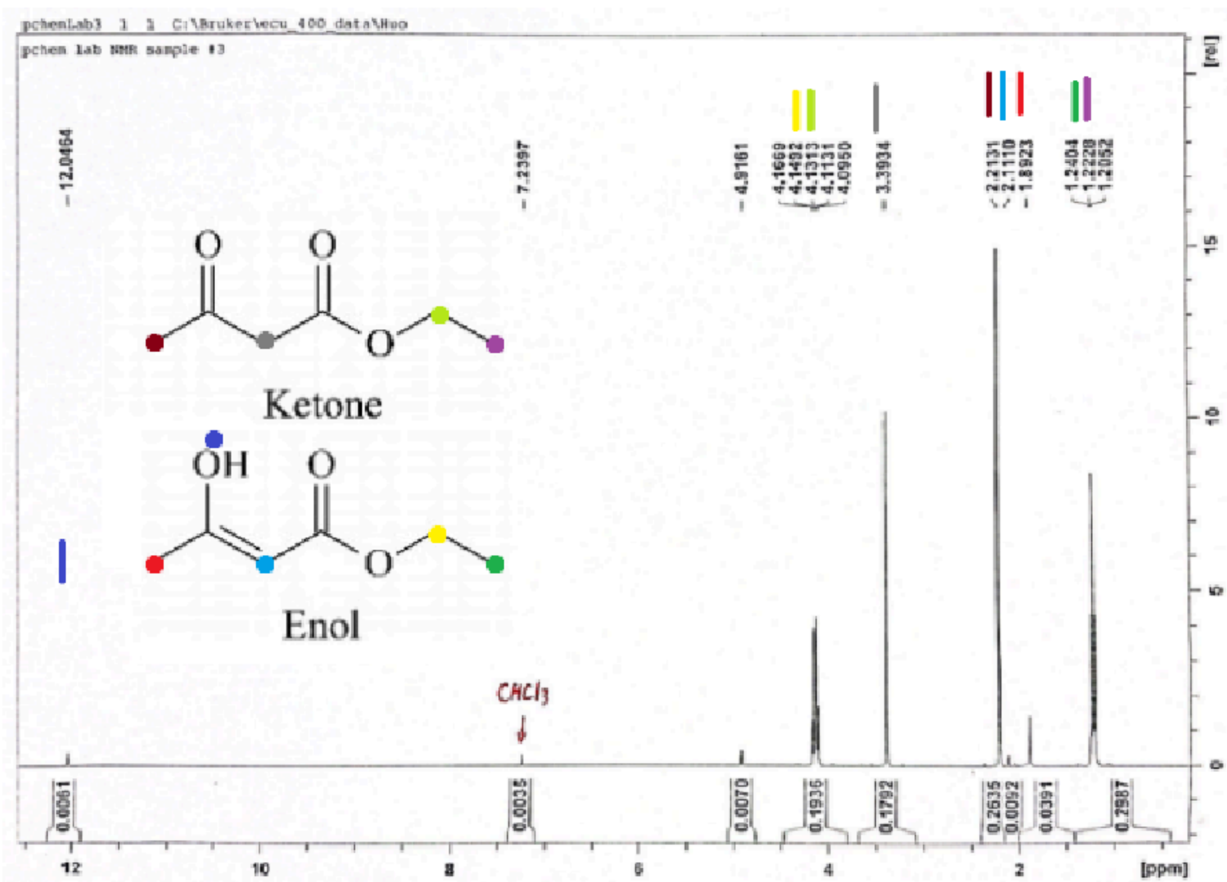

Table 2: NMR data for ethyl acetoacetate

Chemical Shift (ppm)	**Integration Value**	**Splitting Pattern**	**Compound Form**	**Proton Assignment**
1.21-1.24	0.30	Triplet	Keto/Enol	$-OCH_2CH_3$
1.89	0.04	Singlet	Enol	CH_3-C=$C(OH)$-
2.11	0.01	Singlet	Enol	CH_3
2.21	0.26	Singlet	Ketone	CH_3CO-
3.39	0.18	Singlet	Ketone	$-CH_2-$
4.10-4.17	0.19	Quartet	Keto/Enol	$-OCH_2CH_3$
12.05	0.01	Singlet	Enol	$-OH$

CALCULATIONS

The equilibrium constants were calculated using the integration values from Tables 2 and 3. Table 2 corresponds to acetylacetone calculations, and Table 3 corresponds to ethyl acetoacetate calculations. Integration values are proportional to the number of protons producing each signal, when necessary correction factors were applied.

Equilibrium Constant for Acetylacetone

For acetylacetone, four different keto and enol signals were used in the calculation of K_{eq} as specified in the lab manual. The proton assignments are located in Table 1.

Calculation 1: Enol -CH3 vs Keto -CH3

Both signals represent 3 protons, no correction factor needed

$$K_{eq1} = \frac{[enol]}{[keto]} = \frac{12.55}{2.30} = 5.46$$

Calculation 2: Enol =CH- vs Keto -CH2-

The enol =CH- represents one proton, while -CH2- represents two protons. This means the enol integration must be multiplied by two.

$$K_{eq2} = \frac{2[Enol]}{[Keto]} = \frac{2(2.00)}{0.69} = 5.80$$

Calculation 3: Enol -OH vs Keto -CH2-

Both represent different numbers of protons. Multiply enol by 2.

$$K_{eq3} = \frac{2[Enol]}{[Keto]} = \frac{2(1.84)}{0.69} = 5.33$$

Calculation 4: Enol = CH- vs Keto -CH3

The =CH- Enol represents 1, while the Keto -CH3 represents 3. Multiply Enol by 3.

$$K_{eq4} = \frac{3[Enol]}{[Keto]} = \frac{3(2.00)}{2.30} = 2.61$$

Calculation 5: Average of K_{eq}

$$K_{eq}avg = \frac{(K_{eq1}+K_{eq2}+K_{eq3}+K_{eq4})}{4} = \frac{(5.46 + 5.80 + 5.33 + 2.61)}{4} = 4.80$$

Gibbs Free Energy Calculations for Acetylacetone

The Gibbs free energy change was calculated using Equation 3, where $R = 8.314 J mol^{-1} K^{-1}$ at T = 298 Kelvin

$$\Delta G° = - RT ln(K_{eq}) = - (8.314 J mol^{-1} K^{-1})(298\ K) ln(4.80) = - 3.89 kJ\ mol^{-1}$$

Equilibrium Constant for Ethyl Acetoacetate

For ethyl acetoacetate, one calculation utilizing the methyl groups was performed as specified in the lab manual. The proton assignments are located in Table 2.

Calculation 6: Enol =C-CH3 vs Keto -COCH3

Both signals represent 3 protons, so no correction factor is needed.

$$K_{eq} = \frac{[Enol]}{[Keto]} = \frac{0.04}{0.26} = 0.15$$

Calculation 7: Gibbs Free Energy Calculations for Ethyl Acetoacetate

Utilizing Equation 3:

$$\Delta G° = - RTln(K_{eq}) = - (8.314 J mol^{-1} K^{-1})(298\ K) ln(0.15) = 4.7 kJ\ mol^{-1}$$

The calculations indicate that acetylacetone favors the enol tautomer (negative $\Delta G°$), while the ethyl acetoacetate favors the keto tautomer (positive $\Delta G°$).

RESULTS AND DISCUSSION

In this experiment, three compounds were analyzed by ^1H NMR, with ethyl acetate serving as a control for splitting and integration accuracy. Acetylacetone and ethyl acetoacetate were examined to determine their keto-enol equilibrium positions.

In acetylacetone, there is left-right symmetry. This explains why the carbons in the 1,5 positions appear as the same taller singlet for both respective keto and enol tautomers. The carbon in position 3 also appears as a singlet due to having no hydrogen neighbors; the integration value on this will be smaller due to fewer hydrogens present. A prominent indicator that the enol tautomer is present is the small peak heavily deshielded around 15 that indicates that an OH hydrogen is present.

In ethyl acetoacetate, the CH3 triplet integrated to approximately three protons, while the CH2 quartet integrated to two. This is consistent with the 3:2 ratio, indicating that the instrument's integration was reliable enough for keto-enol tautomer identification. The OH was also present in this spectrum, indicating that the enol was present. Identifying the correct set of singlets was more challenging than the acetylacetone due to less of the unfavored tautomer being produced. While the keto peaks were easily identifiable as expected, the enol peaks were much smaller and could very easily be confused with noise.

For acetylacetone, four independent equilibrium constants were obtained after comparing the keto and enol signals. Proton correction factors were applied when needed, and the calculated values ranged from 2.61 to 5.80. The average equilibrium factor was 4.80, compared to a literature value of 5.80; this is a difference of 17%. The Gibbs free energy calculation resulted in $-\ 3.89kJ\ mol^{-1}$, that aligns with the expected enol favored stabilization indicated with the negative number.

Ethyl acetoacetate displayed opposing tautomeric behavior. Only one calculation for an equilibrium constant was used from the methyl group. The average equilibrium factor was 0.15 compared to the literature value of 0.08, yielding a difference of 83%. The Gibbs free energy change was $4.7\ kJ\ mol^{-1}$, which indicates that the keto tautomer is more stable. This is supported by the lack of hydrogen bonding, which would favor the keto form.

Overall, the experimental results support the two tautomeric preferences displayed by the two β-dicarbonyl compounds. Acetylacetone exists as an enol, whereas ethyl acetoacetate exists as a keto. Deviations from these literature values may be due to solvent effects, spectral error, or preparation error. Despite these differences, the conclusion still aligns with the expected result from chemical theory, confirming NMR’s utility in observing and quantifying keto-enol tautomerization.

REFERENCES

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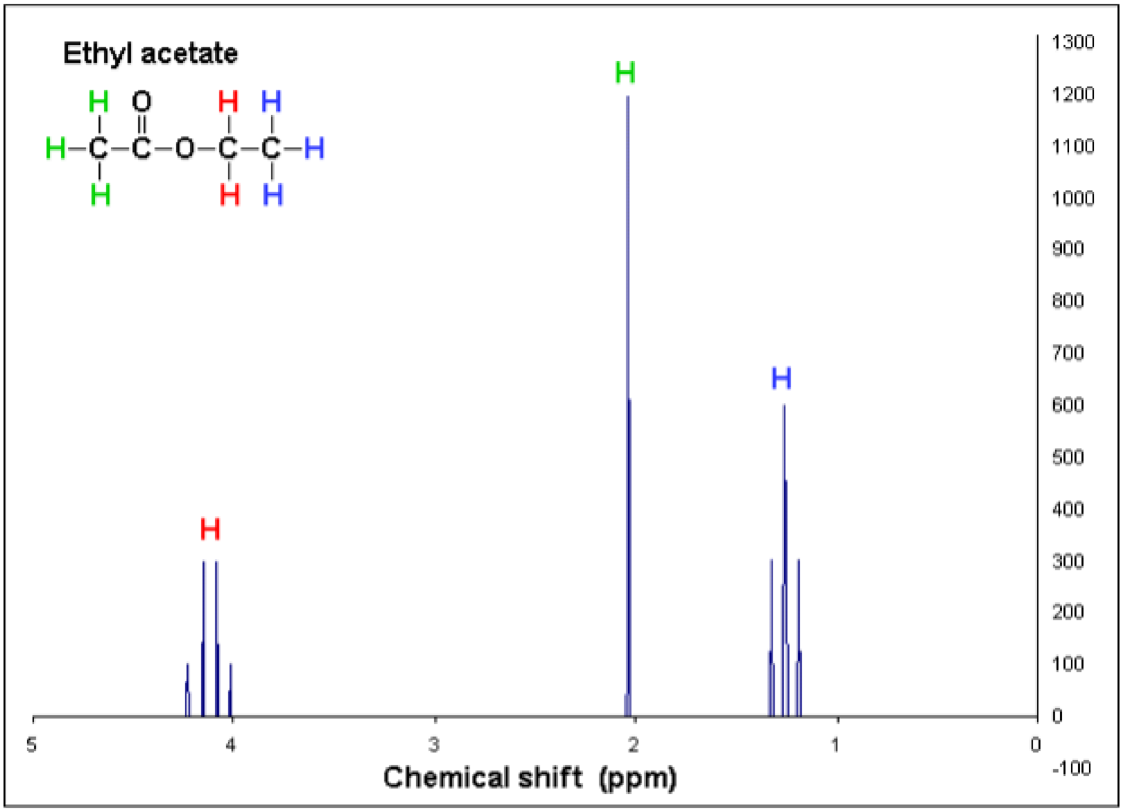
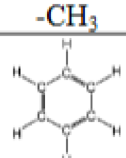
SUPPLEMENTAL INFORMATION

Peak assignment reference figure:



Ethyl Acetate

Table 3: Expected Chemical Shifts

Type of Proton	Chemical Shift (ppm)	
$-CH_3$	0.9	
	7.27	
CH_3CH_2OH	3.59	
CH_3CHO	9.72	

Calculation Example

$$Corrected\ toluene\ signal = 16.71 * \frac{2}{3} = 11.14$$

$$\frac{toluene}{\text{1,1−dibromomethane}} = \frac{11.14}{2.73} = 4.08$$

So, the mole ratio of toluene/1,1-dibromomethane in the mixture is 4.08.